1️⃣ Interpret mathematical outcomes in relation to the situation (Most Relevant)

Relevant item:

5. Interpret mathematical outcomes in a variety of formats in relation to a situation interpret

How it is reflected

The problem presents mathematical information in **tabular form**, including:

- Numerical quantities (2.8 million)
- Percentages (85.8%)

Students must understand that:

- **2.8 million** represents the number of households that own TVs.
- **85.8%** represents the proportion of all households that own TVs.

Using this information, students estimate the total number of households:

$$\text{Total households} = \frac{2.8}{0.858} \approx 3.26 \text{ million}$$

The closest answer is:

C. 3.3 million

Competency demonstrated

Students must:

- Interpret information presented in multiple formats
- Understand the relationship between quantity and percentage
- Relate the mathematical result back to the real-world context

Therefore, this is the **core Interpret competency** in this task.

2️⃣ Understand the relationship between context and mathematical representation

Relevant item:

9. Understand the relationship between the context of the problem and representation of the mathematical solution interpret

How it is reflected

Students must understand the relationship between the real-world context and the mathematical representation.

Context meaning:

- **2.8 million** → number of households that own TVs
- **85.8%** → proportion of all households

Mathematical relationship:

$$\text{part} = \text{percentage} \times \text{total}$$

Thus:

$$2.8 = 0.858 \times T$$

Solving for the total number of households:

$$T = \frac{2.8}{0.858}$$

This reflects the ability to connect **real-world meaning with mathematical representation**.

3️⃣ Identify whether a mathematical result makes sense in context

Relevant item:

2. Identify whether a mathematical result or conclusion makes sense given the context of a problem interpret

How it is reflected

Students must evaluate whether the result is reasonable.

Since **85.8% of households own TVs**, the total number of households should be **slightly larger than 2.8 million**.

Thus a reasonable estimate should be around **3 million**.

Looking at the answer choices:

- 2.4 million → smaller than 2.8 (not reasonable)
- 2.9 million → slightly small
- **3.3 million → reasonable**
- 3.8 million → somewhat large

Students must use contextual reasoning to judge the plausibility of the result.

4️⃣ Use contextual knowledge to judge the reasonableness of results

Relevant item:

6. Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure interpret

How it is reflected

Students can use approximate reasoning:

Since **85.8% ≈ 86%**, then

$$\text{Total} \approx \frac{2.8}{0.86}$$

The total should therefore be **slightly greater than 2.8 million but clearly less than 4 million**.

This demonstrates the use of real-world reasoning to evaluate results.

5️⃣ Construct and communicate explanations in context

Relevant item:

7. Construct and communicate explanations and arguments in the context of the problem interpret

How it is reflected

Students should explain the reasoning:

Since **85.8% of households own TVs**, and **2.8 million represents that portion**, the total number of households is:

$$2.8 \div 0.858 \approx 3.3 \text{ million}$$

This reflects the ability to **explain a mathematical conclusion within the context of the situation**.

1️⃣ Interpret mathematical outcomes in relation to the situation（最相关）

对应条目：

5. Interpret mathematical outcomes in a variety of formats in relation to a situation interpret

体现方式

题目提供的数学信息以 **表格形式**呈现，包括：

- 数量（2.8 million）
- 百分比（85.8%）

学生需要理解：

2.8 million 表示 **拥有电视的家庭数量**，
85.8% 表示 **拥有电视的家庭占全部家庭的比例。**

然后根据这些信息推断：

$$\text{Total households} = \frac{2.8}{0.858} \approx 3.26 \text{ million}$$

最接近答案：

C. 3.3 million

体现能力

学生需要：

- 解读表格中的多种信息形式
- 理解数量与比例之间的关系
- 将数学结果解释回现实情境

因此这是本题**最核心的 Interpret 能力。**

2️⃣ Understand the relationship between context and mathematical representation

对应条目：

9. Understand the relationship between the context of the problem and representation of the mathematical solution interpret

体现方式

学生必须理解：

现实含义：

- 2.8 million → 有电视的家庭数
- 85.8% → 占全部家庭比例

数学关系：

$$\text{part} = \text{percentage} \times \text{total}$$

即：

$$2.8 = 0.858 \times T$$

求：

$$T = \frac{2.8}{0.858}$$

这里体现了：

现实情境 ↔ 数学表达之间的对应关系。

3️⃣ Identify whether a mathematical result makes sense in context

对应条目：

2. Identify whether a mathematical result or conclusion makes sense given the context

interpret

体现方式

学生需要判断：

如果 **85.8% 的家庭拥有电视**，

那么：

总家庭数应该 **略大于 2.8 million。**

因此合理范围应为：

- 大约 **3 million 左右**

选项中：

- 2.4 million → 小于 2.8（不合理）
- 2.9 million → 偏小
- **3.3 million → 合理**
- 3.8 million → 偏大

因此需要利用现实理解来判断答案的合理性。

4️⃣ Use contextual knowledge to judge the reasonableness of results

对应条目：

6. Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure interpret

体现方式

学生需要利用现实逻辑：

如果 **85.8% ≈ 86%**

则：

$$\text{Total} \approx \frac{2.8}{0.86}$$

结果应该：

略大于 2.8 million，但不会远大于 4 million。

这属于利用现实常识对结果进行判断。

5️⃣ Construct and communicate explanations in the context

对应条目：

7. Construct and communicate explanations and arguments in the context of the problem

interpret

体现方式

学生需要解释：

因为：

85.8% 的家庭拥有电视
而 2.8 million 是这一部分

所以：

总家庭数 = 2.8 ÷ 0.858 ≈ 3.3 million

这体现了**在情境中解释数学结果的能力。**